QP CODE: 24900030

MAHATMA GANDHI UNIVERSITY, KOTTAYAM FIRST SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION NOVEMBER 2024
First Semester
Discipline Specific Core Course - MG1DSCBBT101, MG1DSCBTG100 - FUNDAMENTALS FOR BIOTECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Scheme for Valuation

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A Answer the Questions in One Word or Sentence

Answer all questions.

Each question carries 1 mark

1. Identify the organelle is known as power house of the cell? [U] [1, 2]
   Answer: Mitochondria
2. Name the enzyme used in PCR? [K] [1, 2]
   Answer: (Out of syllabus) (1mark can be given for an attempt with question number)
3. What is the causative organism of sleeping sickness? [U] [1, 2]
   Answer: Trypanosoma
4. Give an example of RNA viruses. [U] [1, 2]
   Answer: Any RNA virus
5. Identify the type of immune cell that produces antibodies in response to antigens. [K] [3]
   Answer: B cell
6. Give an example of a structural protein. [U] [1, 2]
   Answer: Any structural protein
7. Which fatty acid is found in fish oils? [U] [1, 2]
   Answer: Omega 3 fatty acid
8. What are the primary sources of dietary lipids? [K] [1, 2]
   Answer: Any source containing lipids
9. Which enzyme relieves the supercoiling ahead of the replication fork? [U] [2]
   Answer: Topoisomerase/ Gyrase
10. Give an example for plasmid vector. [U] [2]
    Answer: Any vector

(10 × 1 = 10)

Part B Short Answer Questions Answer any 4 questions.

Each question carries 3 marks

11. Give an outline of major milestones in biotechnology. [K] [1, 2]

    Answer: Important historical events (minimum 3 points)

12. Explain the difference between ceramides and sphingomylin. [U] [1, 2]

    Answer: (Out of syllabus) (3 marks can be given for an attempt with question number)

13. How are protozoa transmitted from one host to another? [U] [1, 2]

    Answer: Modes of transmission- 3 points- 3 marks

14. Define prions. [U] [1, 2]

    Answer: (out of syllabus) (3 marks can be given for an attempt with question number)

15. What is the start codon, and which amino acid does it code for? [U] [2]

    Answer: AUG - 2 marks, Methionine- 1 mark

16. Explain the difference between monounsaturated and polyunsaturated fatty acids. [U] [1, 2]

    Answer: Difference (minimum 3 points) - 3 marks

(4 × 3 = 12)

Part C

Short Essay Questions Answer any 3 questions.

Each question carries 6 marks.

17. List out the major application of biotechnology and the career opportunity to the field of Biotechnology [K] [1, 2]

    Answer: Major applications- (minimum 3 points) - 3 marks

    Career opportunities (minimum 3 points) - 3 marks

18. Explain the replication of a virus. [U] [1, 2]

    Answer: Introduction- 2 marks, Life cycle (mention lytic and lysogenic) - 4 marks

19. Differentiate between monohybrid and dihybrid crosses. [U] [1, 2]

    Answer: Monohybrid (with example)- 3 marks

    Dihybrid (with example) - 3 marks

20. Recite the differences between diasaccharides and polysaccharides. [U] [1, 2]

    Answer: Disacchrides - definition- 1 mark, Polysaccharides- definition- 1 mark

    Difference (4 points minimum) - 4 marks

21. Tabulate the difference between prokaryotic and eukaryotic transcription [K] [2]

    Answer: Transcription- introduction- 2 marks, steps- 2 marks, difference- 2 marks

(3 × 6 = 18)

Part D

Essay Questions - Answer any 1 question (10 marks)

22. Explain the key components of the immune system and their functions. [An] [3]

    Answer: Immune system- introduction-2 marks, cells in immune system- 4 marks,

    Functions- 4 marks

23. Discuss with diagrams the steps involved in the process of replication of DNA. [U] [2]

    Answer: Replication- introduction- 2 marks, steps- 5 marks, diagram- 3 marks

(1 × 10 = 10)

END OF THE QUESTION PAPER ***